

## Research Interests

• Nonlinear control • Model predictive control • Learning based control • Modelling & control of energy conversion systems

## Professional Experience

- 2023 – . . . .  **Tenured Associate Professor**, Department of Electrical & Computer Engineering, COMSATS University Islamabad (CUI), Islamabad, Pakistan
- Teaching and student supervision
  - Head of [Control and Energy Systems Research Group \(CESRG\)](#)
  - Convener of folder management and evaluation committee
- 2014 – 2023  **Assistant Professor**, Department of Electrical & Computer Engineering, CUI
- Teaching at graduate & undergraduate levels
  - Supervision/Co-supervision of MS and Ph.D. theses and undergraduate projects
  - Head of CESRG
- 2020 – 2021  **Doctoral Researcher**, Department of Electrical & Computer Engineering, University of Porto, Porto, Portugal  
**Research Project:** *FCT UPWIND– A Multi-Kite System to Harvest High Altitude Wind Power*
- Modelling and parametrization of the induction machine (IM)
  - Development of observer based rotor flux-oriented control of the IM
  - Development of a robust cascade control of the ground station module of the airborne wind energy system (AWES) in all operational phases
  - Development of the supervisory controller for the AWES
- 2015 – 2016  **Visiting Scholar**, Department of Electrical & Computer Engineering, The Ohio State University, Columbus, OH, USA
- Development of a simplified time domain model of underground coal gasification (UCG) process
  - Development of the model based control of the UCG process based on Sliding Mode Control theory
  - Implementation of the developed controller on the actual process model
  - The stability of the zero dynamics, which guarantees the overall stability of the closed loop system
- 2012 – 2014  **Professional Researcher**, Control & Signal Processing Research Group, Capital University of Science & technology, Islamabad, Pakistan  
**Research Project:** *ICT R & D Funds– Simulation and Control of the UCG process*
- Development of the one-dimensional packed bed reactor model for Thar gasifier
  - Numerical solution of the developed model
  - Computer simulation of the numerical solution techniques
  - Parameter estimation for the UCG process
  - Model validation with actual field trials
  - Robust control system design for obtaining a desired calorific value of the product gas mixture

## Professional Experience (continued)

- 2008 – 2012  **Lecturer**, Department of Electrical & Computer Engineering, CUI
- Teaching at undergraduate level
  - Supervision of undergraduate projects
  - Designed and conducted labs for Electric Machines and Control System
- 2007 – 2008  **Design Engineer**, Public Sector Organization, Islamabad, Pakistan
- Integration and testing of electronic modules

## Education

- 2012 – 2016  **Ph.D., Electrical Engineering, COMSATS University Islamabad.**  
Thesis title: [Modeling and Control of Underground Coal Gasification](#).  
CGPA: 3.93/4.00
- 2009 – 2011  **M.Sc., Computer Engineering, University of Engineering & Technology Taxila.**  
Thesis title: Dynamic Modeling and Nonlinear Controller Design for UCG.  
CGPA: 3.83/4.00
- 2003 – 2006  **BS., Electrical Engineering (with honours), University of Engineering & Technology Taxila.**  
Marks: 80.69%

## Funded Projects

-  **Design and Development of Intelligent Battery Pack for Range Extension of Electric Scooter**  
*PSF-TUBITAK Competitive Research Grant 2024, PI: Ali Arshad Uppal, Funding amount: PKR 19.37 Million*
-  **Optimal Control of Active Cell Balancing Network for Lithium-ion Battery Pack**  
*NESCOM (RAC), Pakistan, PI: Ali Arshad Uppal, Funding amount: PKR 0.4 Million*

## Publications

### Journal Articles

- 1 A. Ahmed, **A. A. Uppal**, and Q. Ahmed, "Electric vehicle range extension using nonlinear mpc based active battery cell balancing," *Journal of Process Control*, vol. 164, p. 103 758, 2026, ISSN: 0959-1524.  DOI: <https://doi.org/10.1016/j.jprocont.2026.103758>.
- 2 M. Shakeel, **A. A. Uppal**, N. Tasneem, and Y. Alsmadi, "Optimization of distributed energy resources in distribution networks using multi-objective archimedes optimization algorithm," *Symmetry*, vol. 18, no. 1, 2026, ISSN: 2073-8994.  DOI: [10.3390/sym18010075](https://doi.org/10.3390/sym18010075).
- 3 M. Azmat Ullah, A. Ahmed, **A. A. Uppal**, and Q. Ahmed, "Nmpc-based performance evaluation of active balancing networks of li-ion batteries for overnight ev charging," *IEEE Access*, vol. 13, pp. 136 016–136 026, 2025.  DOI: [10.1109/ACCESS.2025.3594129](https://doi.org/10.1109/ACCESS.2025.3594129).
- 4 S. B. Javed, **A. A. Uppal**, M. R. Azam, and Q. Ahmed, "Model-based quantitative analysis of power losses aware active cell balancing networks with load," *IEEE Transactions on Transportation Electrification*, pp. 1–1, Sep. 2024.  DOI: [10.1109/TTE.2024.3455767](https://doi.org/10.1109/TTE.2024.3455767).
- 5 Y. CW., S. Riaz, **A. A. Uppal**, and J. Iqbal, "Fuzzy fault-tolerant controller with guaranteed performance for mimo systems under uncertain initial state," *International Journal of Control, Automation and Systems*, vol. 22, pp. 2038–2054, Jun. 2024.  DOI: [10.1007/s12555-023-0327-5](https://doi.org/10.1007/s12555-023-0327-5).
- 6 **A. A. Uppal**, S. B. Javed, and Q. Ahmed, "Power losses aware nonlinear model predictive control design for active cell balancing," *IEEE Control Systems Letters*, vol. 7, pp. 3705–3710, Dec. 2023.  DOI: [10.1109/LCSYS.2023.3342550](https://doi.org/10.1109/LCSYS.2023.3342550).

- 7 A. Ahmed, **A. A. Uppal**, and S. B. Javed, "Nonlinear-control-oriented modeling of the multi-variable underground coal gasification process for ucg project thar: A machine learning perspective," *Journal of Process Control*, vol. 131, p. 103 090, Nov. 2023, ISSN: 0959-1524.  DOI: [10.1016/j.jprocont.2023.103090](https://doi.org/10.1016/j.jprocont.2023.103090).
- 8 **A. A. Uppal**, M. R. Azam, and J. Iqbal, "Sliding mode control in dynamic systems," *Electronics*, vol. 12, no. 13, Jul. 2023.  DOI: [10.3390/electronics12132970](https://doi.org/10.3390/electronics12132970).
- 9 A. Ahmed, S. B. Javed, **A. A. Uppal**, and J. Iqbal, "Development of cavlab—a control-oriented matlab based simulator for an underground coal gasification process," *Mathematics*, vol. 11, no. 11, May 2023, ISSN: 2227-7390.  DOI: [10.3390/math11112493](https://doi.org/10.3390/math11112493).
- 10 U. Javaid, A. Mehmood, J. Iqbal, and **A. A. Uppal**, "Neural network and ured observer based fast terminal integral sliding mode control for energy efficient polymer electrolyte membrane fuel cell used in vehicular technologies," *Energy*, vol. 269, p. 126 717, Apr. 2023, ISSN: 0360-5442.  DOI: [10.1016/j.energy.2023.126717](https://doi.org/10.1016/j.energy.2023.126717).
- 11 S. Ahmad, **A. A. Uppal**, M. R. Azam, and J. Iqbal, "Chattering free sliding mode control and state dependent kalman filter design for underground gasification energy conversion process," *Electronics*, vol. 12, no. 4, Feb. 2023, ISSN: 2079-9292.  DOI: [10.3390/electronics12040876](https://doi.org/10.3390/electronics12040876).
- 12 I. U. Rehman, S. B. Javed, A. M. Chaudhry, M. R. Azam, and **A. A. Uppal**, "Model-based dynamic sliding mode control and adaptive kalman filter design for boiler-turbine energy conversion system," *Journal of Process Control*, vol. 116, pp. 221–233, Aug. 2022, ISSN: 0959-1524.  DOI: [10.1016/j.jprocont.2022.06.006](https://doi.org/10.1016/j.jprocont.2022.06.006).
- 13 U. Javaid, J. Iqbal, A. Mehmood, and **A. A. Uppal**, "Performance improvement in polymer electrolytic membrane fuel cell based on nonlinear control strategies—a comprehensive study," *PLOS ONE*, vol. 17, no. 2, pp. 1–20, Mar. 2022.  DOI: [10.1371/journal.pone.0264205](https://doi.org/10.1371/journal.pone.0264205).
- 14 M. Khattak, **A. A. Uppal**, Q. Khan, A. I. Bhatti, Y. M. Alsmadi, V. I. Utkin, and I. Chairez, "Neuro-adaptive sliding mode control for underground coal gasification energy conversion process," *International Journal of Control*, vol. 95, no. 9, pp. 2337–2348, Feb. 2022.  DOI: [10.1080/00207179.2021.1909745](https://doi.org/10.1080/00207179.2021.1909745).
- 15 H. Muazzam, M. K. Ishak, A. Hanif, **A. A. Uppal**, A. Bhatti, and N. A. M. Isa, "Virtual sensor using a super twisting algorithm based uniform robust exact differentiator for electric vehicles," *Energies*, vol. 15, no. 5, Feb. 2022.  DOI: [10.3390/en15051773](https://doi.org/10.3390/en15051773).
- 16 S. Bano, M. R. Azam, **A. A. Uppal**, S. B. Javed, and A. I. Bhatti, "Robust p53 recovery using chattering free sliding mode control and a gain-scheduled modified utkin observer," *Journal of Theoretical Biology*, vol. 532, p. 110 914, Jan. 2022, ISSN: 0022-5193.  DOI: [10.1016/j.jtbi.2021.110914](https://doi.org/10.1016/j.jtbi.2021.110914).
- 17 S. B. Javed, V. I. Utkin, **A. A. Uppal**, R. Samar, and A. I. Bhatti, "Data-driven modeling and design of multivariable dynamic sliding mode control for the underground coal gasification project thar," *IEEE Transactions on Control Systems Technology*, vol. 30, no. 1, pp. 153–165, Jan. 2022.  DOI: [10.1109/TCST.2021.3057633](https://doi.org/10.1109/TCST.2021.3057633).
- 18 **A. A. Uppal**, M. C. R. M. Fernandes, S. Vinha, and F. A. C. C. Fontes, "Cascade control of the ground station module of an airborne wind energy system," *Energies*, vol. 14, no. 24, Dec. 2021, ISSN: 1996-1073.  DOI: [10.3390/en14248337](https://doi.org/10.3390/en14248337).
- 19 M. Riaz, A. R. Yasin, **A. A. Uppal**, and A. Yasin, "A novel dynamic integral sliding mode control for power electronic converters," *Science Progress*, vol. 104, no. 4, p. 00 368 504 211 044 848, Oct. 2021.  DOI: [10.1177/00368504211044848](https://doi.org/10.1177/00368504211044848).
- 20 A. M. Chaudhry, **A. A. Uppal**, and S. Bram, "Model predictive control and adaptive kalman filter design for an underground coal gasification process," *IEEE Access*, vol. 9, pp. 130 737–130 750, Sep. 2021.  DOI: [10.1109/ACCESS.2021.3114260](https://doi.org/10.1109/ACCESS.2021.3114260).
- 21 A. Mohsin, Y. Alsmadi, **A. A. Uppal**, and S. M. Gulfam, "A modified simplex based direct search optimization algorithm for adaptive transversal fir filters," *Science Progress*, vol. 104, no. 2, p. 00 368 504 211 025 409, Jun. 2021.  DOI: [10.1177/00368504211025409](https://doi.org/10.1177/00368504211025409).
- 22 Y. M. Alsmadi, I. U. Rehman, **A. A. Uppal**, V. Utkin, I. Chairez, and M. Ibbini, "Super-twisting-based sliding mode control of drum boiler energy conversion systems," *International Journal of Control*, vol. 95, no. 7, pp. 1888–1897, Feb. 2021.  DOI: [10.1080/00207179.2021.1884293](https://doi.org/10.1080/00207179.2021.1884293).

- 23 S. B. Javed, A. A. Uppal, R. Samar, and A. I. Bhatti, "Design and implementation of multi-variable  $H_\infty$  robust control for the underground coal gasification project thar," *Energy*, vol. 216, p. 119 000, Feb. 2021, ISSN: 0360-5442.  DOI: [10.1016/j.energy.2020.119000](https://doi.org/10.1016/j.energy.2020.119000).
- 24 Q. Irum, S. A. Khan, A. A. Uppal, and L. Krivodonova, "Galerkin finite element based modeling of one dimensional packed bed reactor for underground coal gasification (ucg) process," *IEEE Access*, vol. 8, pp. 223 130–223 139, Dec. 2020.  DOI: [10.1109/ACCESS.2020.3044194](https://doi.org/10.1109/ACCESS.2020.3044194).
- 25 U. Javaid, A. Mehmood, A. A. Uppal, F. Imtiaz, and J. Iqbal, "Operational efficiency improvement of pem fuel cell—a sliding mode based modern control approach," *IEEE Access*, vol. 8, pp. 95 823–95 831, May 2020.  DOI: [10.1109/ACCESS.2020.2995895](https://doi.org/10.1109/ACCESS.2020.2995895).
- 26 M. Ilyas, J. Iqbal, S. Ahmad, A. A. Uppal, W. A. Imtiaz, and R. A. Riaz, "Hypnosis regulation in propofol anaesthesia employing super-twisting sliding mode control to compensate variability dynamics," *IET Systems Biology*, vol. 14, no. 2, pp. 59–67, Apr. 2020.  DOI: [10.1049/iet-syb.2018.5080](https://doi.org/10.1049/iet-syb.2018.5080).
- 27 M. Rizwan Azam, V. I. Utkin, A. A. Uppal, and A. I. Bhatti, "Sliding mode controller–observer pair for p53 pathway," *IET Systems Biology*, vol. 13, no. 4, pp. 204–211, Jul. 2019.  DOI: [10.1049/iet-syb.2018.5121](https://doi.org/10.1049/iet-syb.2018.5121).
- 28 Y. M. Alsmadi, A. M. Abdel-hamed, A. E. Ellissy, A. S. El-Wakeel, A. Y. Abdelaziz, V. Utkin, and A. A. Uppal, "Optimal configuration and energy management scheme of an isolated micro-grid using cuckoo search optimization algorithm," *Journal of the Franklin Institute*, vol. 356, no. 8, pp. 4191–4214, May 2019, ISSN: 0016-0032.  DOI: [10.1016/j.jfranklin.2018.12.014](https://doi.org/10.1016/j.jfranklin.2018.12.014).
- 29 S. B. Javed, A. A. Uppal, A. I. Bhatti, and R. Samar, "Prediction and parametric analysis of cavity growth for the underground coal gasification project thar," *Energy*, vol. 172, pp. 1277–1290, Apr. 2019, ISSN: 0360-5442.  DOI: [10.1016/j.energy.2019.02.005](https://doi.org/10.1016/j.energy.2019.02.005).
- 30 A. R. Yasin, M. Ashraf, A. I. Bhatti, and A. A. Uppal, "Fixed frequency sliding mode control of renewable energy resources in dc micro grid," *Asian Journal of Control*, vol. 21, no. 4, pp. 2074–2086, Mar. 2019.  DOI: [10.1002/asjc.2057](https://doi.org/10.1002/asjc.2057).
- 31 A. A. Uppal, S. S. Butt, Q. Khan, and H. Aschemann, "Robust tracking of the heating value in an underground coal gasification process using dynamic integral sliding mode control and a gain scheduled modified utkin observer," *Journal of Process Control*, vol. 73, pp. 113–122, Jan. 2019, ISSN: 0959-1524.  DOI: [10.1016/j.jprocont.2018.11.005](https://doi.org/10.1016/j.jprocont.2018.11.005).
- 32 A. M. Chaudhry, A. A. Uppal, Y. M. Alsmadi, A. I. Bhatti, and V. I. Utkin, "Robust multi-objective control design for underground coal gasification energy conversion process," *International Journal of Control*, vol. 93, no. 2, pp. 328–335, Aug. 2018.  DOI: [10.1080/00207179.2018.1516893](https://doi.org/10.1080/00207179.2018.1516893).
- 33 A. A. Uppal, Y. M. Alsmadi, V. I. Utkin, A. I. Bhatti, and S. A. Khan, "Sliding mode control of underground coal gasification energy conversion process," *IEEE Transactions on Control Systems Technology*, vol. 26, no. 2, pp. 587–598, Apr. 2017.  DOI: [10.1109/TCST.2017.2692718](https://doi.org/10.1109/TCST.2017.2692718).
- 34 I. Khan, A. I. Bhatti, A. A. Uppal, and Q. Khan, "Robustness and performance parameterization of smooth second order sliding mode control," *International Journal of Control, Automation and Systems*, vol. 14, no. 3, pp. 681–690, Jun. 2016, ISSN: 2005-4092.  DOI: [10.1007/s12555-014-0181-6](https://doi.org/10.1007/s12555-014-0181-6).
- 35 A. A. Uppal, A. I. Bhatti, E. Aamir, R. Samar, and S. A. Khan, "Optimization and control of one dimensional packed bed model of underground coal gasification," *Journal of Process Control*, vol. 35, pp. 11–20, Nov. 2015, ISSN: 0959-1524.  DOI: [10.1016/j.jprocont.2015.08.002](https://doi.org/10.1016/j.jprocont.2015.08.002).
- 36 A. A. Uppal, A. I. Bhatti, E. Aamir, R. Samar, and S. A. Khan, "Control oriented modeling and optimization of one dimensional packed bed model of underground coal gasification," *Journal of Process Control*, vol. 24, no. 1, pp. 269–277, Jan. 2014, ISSN: 0959-1524.  DOI: [10.1016/j.jprocont.2013.12.001](https://doi.org/10.1016/j.jprocont.2013.12.001).

## Conference Proceedings

- 1 A. A. Uppal, M. R. Azam, A. Ahmed, and Q. Ahmed, "Design and evaluation of nonlinear mpc for active load balancing in adjacent battery cells□this work was not supported by any organization.," in *13th IFAC Symposium on Nonlinear Control Systems NOLCOS 2025*, 2025, pp. 460–465.  DOI: <https://doi.org/10.1016/j.ifacol.2025.11.077>.

- 2 A. Ahmed, **A. A. Uppal**, Q. Ahmed, and M. Rizwan Azam, "Performance analysis of model predictive controllers for active cell balancing of electric vehicle battery," in *2024 21st International Bhurban Conference on Applied Sciences and Technology (IBCAST)*, 2024, pp. 288–293.  DOI: [10.1109/IBCAST61650.2024.10876938](https://doi.org/10.1109/IBCAST61650.2024.10876938).
- 3 M. R. Azam, A. Ahmed, **A. A. Uppal**, and Q. Ahmed, "Nonlinear model predictive control design for active cell balancing and thermal management," in *2024 IEEE Conference on Control Technology and Applications (CCTA)*, Aug. 2024, pp. 46–51.  DOI: [10.1109/CCTA60707.2024.10666603](https://doi.org/10.1109/CCTA60707.2024.10666603).
- 4 **A. A. Uppal**, S. B. Javed, and Q. Ahmed, "Power losses aware nonlinear model predictive control design for active cell balancing," in *American Control Conference 2024*, Jul. 2024.
- 5 S. B. Javed, **A. A. Uppal**, M. R. Azam, K. Shehzad, and Q. Ahmed, "Model-based quantitative analysis of a capacitive cell balancing technique using soc estimator," in *2022 IEEE Conference on Control Technology and Applications (CCTA)*, Dec. 2022, pp. 670–675.  DOI: [10.1109/CCTA49430.2022.9966110](https://doi.org/10.1109/CCTA49430.2022.9966110).
- 6 **A. A. Uppal**, S. S. Butt, A. I. Bhatti, and H. Aschemann, "Integral sliding mode control and gain-scheduled modified utkin observer for an underground coal gasification energy conversion process," in *2018 23rd International Conference on Methods and Models in Automation and Robotics (MMAR)*, Oct. 2018, pp. 357–362.  DOI: [10.1109/MMAR.2018.8486053](https://doi.org/10.1109/MMAR.2018.8486053).
- 7 G. Murtaza, A. I. Bhatti, Q. Ahmed, and **A. A. Uppal**, "Nonlinear robust control of atkinson cycle engine," in *20th IFAC World Congress*, Jul. 2017, pp. 3685–3690.  DOI: [10.1016/j.ifacol.2017.08.562](https://doi.org/10.1016/j.ifacol.2017.08.562).
- 8 M. Azam, A. Bhatti, **A. A. Uppal**, and M. Babar, "Sensitivity analysis of wnt signaling pathway," in *Proceedings of 2013 10th International Bhurban Conference on Applied Sciences & Technology (IBCAST)*, May 2013, pp. 122–127.  DOI: [10.1109/IBCAST.2013.6512143](https://doi.org/10.1109/IBCAST.2013.6512143).
- 9 **A. A. Uppal**, A. I. Bhatti, R. Samar, Q. Ahmed, and E. Aamir, "Model development of ucg and calorific value maintenance via sliding mode control," in *2012 International Conference on Emerging Technologies*, Oct. 2012, pp. 1–6.  DOI: [10.1109/ICET.2012.6375477](https://doi.org/10.1109/ICET.2012.6375477).

## Student Supervision

### ■ MS Students

1. Mr. Imran Ahmed, COMSATS University Islamabad (2021–2023)  
Supervisor: Dr. Rizwan Azam, co-supervisor: **Dr. Ali Arshad Uppal**
2. Mr. Afaq Ahmed, COMSATS University Islamabad (2021–2023)  
Supervisor: **Dr. Ali Arshad Uppal** co-supervisor: Dr. Syed Bilal Javed
3. Ms. Sara Sarfraz, COMSATS University Islamabad (2019–2021)  
Supervisor: **Dr. Ali Arshad Uppal**, co-supervisor: Dr. Qudrat Khan
4. Ms. Shehar Bano, COMSATS University Islamabad (2018–2020)  
supervisor: **Dr. Ali Arshad Uppal**, co-supervisor: Dr. Rizwan Azam
5. Ms. Fajar Mukhtar, COMSATS University Islamabad (2018–2020)  
supervisor: **Dr. Ali Arshad Uppal**, co-supervisor: Dr. Syed Bilal Javed
6. Mr. Sohail Ahmed, COMSATS University Islamabad (2018–2020)  
Supervisor: **Dr. Ali Arshad Uppal**
7. Mr. Bilal Arif, COMSATS University Islamabad (2018–2020)  
Supervisor: prof. Shahid A. Khan, co-supervisor: **Dr. Ali Arshad Uppal**
8. Mr. Mutahir, COMSATS University Islamabad (2016–2018)  
supervisor: prof. Shahid A. Khan, co-supervisor: **Dr. Ali Arshad Uppal**
9. Mr. Imtiaz Ur Rehman, COMSATS University Islamabad (2016–2018)  
Supervisor: prof. Shahid A. Khan, co-supervisor: **Dr. Ali Arshad Uppal**
10. Mr. Fahad Imtiaz, COMSATS University Islamabad (2014–2016)  
Supervisor: Dr. Adeel Mehmood, co-supervisor: **Dr. Ali Arshad Uppal**

### ■ Ph.D. students

1. Ms. Shehar Bano, COMSATS University Islamabad (2024–present)  
Supervisor: **Dr. Ali Arshad Uppal**
2. Mr. Muhammad Shakeel, COMSATS University Islamabad (2019–2026)  
Supervisor: **Dr. Ali Arshad Uppal**
3. Mr. Azmat Ullah, COMSATS University Islamabad (2019–2026)  
Supervisor: **Dr. Ali Arshad Uppal**, co-supervisor: Dr. Rizwan Azam
4. Dr. Usman Javed, COMSATS University Islamabad (2016–2023)  
Supervisor: **Dr. Ali Arshad**, co-supervisor: Dr. Adeel Mehmood
5. Dr. Qudsiya Irum, COMSATS University Islamabad (2016–2022)  
Supervisor: Prof. Shahid A. Khan, co-supervisor: **Dr. Ali Arshad Uppal**
6. Dr. Syed Bilal Javed, Capital University of Science & Technology, Islamabad (2016–2021)  
Supervisor: Dr. Raza Samar, co-supervisor: **Dr. Ali Arshad Uppal**



## Teaching Experience

### Graduate courses at COMSATS University Islamabad

- Intelligent Control Systems (ECI761) in Spring 2022
- Nonlinear Systems and Controls (ECI760) in Fall 2021 and Fall 2022
- Linear Control Systems (ECI660) in Fall 2019 and Fall 2024
- Linear Systems Theory (ECI665) in Spring (2017, 2018 and 2019) and Fall 2017
- Robust Control Systems (ECI765) in Fall 2018

### Undergraduate courses at COMSATS University Islamabad

- Control Systems (EEE325, CPE325) in Fall 2023, Spring 2023, and from 2011–2021
- Electric Machines (EEE371) in Spring 2023, and from 2008–2011
- Electronics-I (EEE231) in Fall 2022

## Workshops& Presentations

### Workshops

1. **Battery Management Systems for Electric Mobility: Modeling, Control, Theory and Implementation**, at the department of Electrical Engineering, COMSATS University Islamabad, 13–14 July, 2026.
2. **Predictive Control with Machine Learning–A Practical Approach**, at the department of Electrical and Computer Engineering, COMSATS University Islamabad, 18–20 November, 2024.
3. **L<sup>A</sup>T<sub>E</sub>X–The Way of Scientific Writing**, at the department of Electrical and Computer Engineering, COMSATS University Islamabad on 17 October 2022
4. **Typesetting in L<sup>A</sup>T<sub>E</sub>X**, at Capital University of Science & Technology, Islamabad on March 13 2018, July 17 2019 and August 5 2021, respectively

### Presentations

1. **Optimal Control of Active Cell Balancing for Electric Vehicle Batteries**, in 3<sup>rd</sup> International Faculty Development Program on "Advanced Topics in Control Systems" organized by Department of Electrical Engineering of K.K.Wagh Institute of Engineering Education and Research, Nashik, India, 19<sup>th</sup> October, 2024
2. **Model Predictive Control Design for Targeted Drug Therapy to Recover p53 in Cancer Treatment**, in Portuguese Meeting on Optimal Control, 2021
3. **AI Based Control of Industrial Processes**, in IEEE 2021 International Conference on Digital Futures and Transformative Technologies, SEECS, NUST, Islamabad, Pakistan, 2021 (*Guest speaker*)
4. **Sliding mode control of the underground coal gasification energy conversion process**, in Symposium on Control Systems, SEECS, NUST 2019 (*Guest speaker*)
5. **Development of a simplified control-oriented model of the underground coal gasification process**, at Department of Electrical & Computer Engineering, The Ohio State University, Columbus, OH, USA, 2015

## Computer Skills

- |                   |                                   |             |
|-------------------|-----------------------------------|-------------|
| • MATLAB/Simulink | • C++                             | • Maple     |
| • Python          | • L <sup>A</sup> T <sub>E</sub> X | • MS Office |

## Awards & Achievements

- 2025 ■ **Recipient of TÜBİTAK-Pakistan Science Foundation (PSF) Joint Support Program 2024 Grant**–Awarded funding for the project “Design and Development of Intelligent Battery Pack for Range Extension of Electric Scooter”, **which was ranked number 1 among the top nine selected projects in the prestigious program.**
- 2022 ■ **Senior Member IEEE**
- 2020–2021 ■ **Postdoctoral Scholarship** from FEUP, University of Porto, Portugal
- 2015 ■ **International Research Support Initiative Program Scholarship** from higher education commission, Pakistan
- 2014–2015 ■ **Research Productivity Award** from COMSATS University Islamabad
- 2012–2013 ■ **Ph.D. Scholarship** from UCG project Thar, Pakistan
- 2012–2016 ■ **In-house Ph.D. Scholarship** from COMSATS University Islamabad, Pakistan
- 2010–2011 ■ **MS Scholarship** from UCG project Thar, Pakistan
- **Merit MS Scholarship** from Center of Advanced Studies in Engineering, University of Engineering & Technology, Taxila, Pakistan

## Community Services

- 2022–2023 ■ **Guest Editor** in the special issue *Sliding Model Control in Dynamic Systems*, MDPI Electronics.
- 2016–... ■ **Academic Reviewer** for the Journal of process control, IEEE Control Systems Letters, IEEE Transactions on (Industrial Informatics, Mechatronics, Transportation & Electrification), International Journal of Hydrogen Energy, Energy Conversion and Management, American Control Conference, IEEE Access, and various journals of MDPI.

## References

**Name** Dr. Aamer Iqbal Bhatti  
**Position** Professor  
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